

CARMA: Online Cluster-Adaptive Eviction for GPTCache

Cluster-Adaptive Reuse and Miss-pressure Admission

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Executive result. I added an opt-in, CPU-only eviction policy to GPTCache that learns bounded semantic topics and paraphrase cells online, adapts their quotas with decayed demand and miss pressure, and rejects weak one-shot admissions. The implementation preserves the existing LRU default and aligns SQLite and FAISS deletion through vector-first cleanup, targeted rollback, and fail-closed reconciliation. On ten paired 10,000-request seeds at capacity 100, CARMA improved category-shift valid-hit rate by **4.885 percentage points** over the per-seed stronger LRU/LFU baseline (95% paired bootstrap CI [4.677, 5.088], Holm-adjusted $p = 0.0234$). It also passed stationary non-regression and token-saving gates. Five fresh SQLite/FAISS processes per policy showed worst ratios across seeds of $1.032\times$ LRU for post-embedding cache-path p95 latency, $0.997\times$ LRU for throughput, and $1.043\times$ LRU for sampled peak RSS.

The preregistered scan-return gate **failed**: CARMA returned 30,000/30,000 valid hits, but LFU already returned 29,998/30,000, leaving only +0.0067 percentage points rather than the required +5.0. This report retains that negative result and distinguishes measured evidence from protocol deviations.

Software baseline	GPTCache commit <code>c59fb3a6152a4458b2a070ca183b61c4b614095f</code>
Primary paper	SCALM, arXiv:2406.00025v1
Feature branch	<code>feature/online-cluster-aware-cache</code>
Primary policy config	topic .70, cell .97, half-life 500, quota strength 1.0

1 Introduction and related work

Semantic caches store responses under vector representations of queries and reuse an answer when a later query is sufficiently similar. GPTCache provides a modular open-source implementation: embeddings, vector search, similarity evaluation, scalar/vector storage, and eviction can be configured separately [1]. At the selected commit, the in-memory manager offers global LRU, LFU, FIFO, and random replacement over opaque row identifiers. Those policies know recency or frequency, but not whether several residents cover one redundant paraphrase region or whether a miss belongs to a growing topic.

SCALM motivates semantic structure as a cache-management signal. It analyzes MOSS and LMSYS conversations, derives hierarchical semantic patterns, ranks them using hit/token-saving evidence, and reports large relative gains over GPTCache baselines [2]. Its pattern discovery is corpus-oriented. The question here is deliberately different:

Can a bounded policy discover useful semantic structure *during* the request stream, without offline ranking, a learned model, token cost as an input, or a paid LLM call?

This question leads to three engineering requirements. First, answer correctness must remain controlled by GPTCache’s retrieval threshold; a policy cluster may guide retention but may never declare a response correct. Second, all state must remain bounded by cache capacity rather than trace length. Third, eviction must remove scalar and vector records together so a rejected FAISS vector cannot shadow a valid neighbor at `top_k=1`.

Contribution and claim boundary

The contribution is CARMA, a deterministic online policy plus backward-compatible GPTCache integration, failure-safe cleanup, restart reconstruction, tests, and a reproducible evaluation pipeline. The results support an adaptive phase-shift benefit; every individual system diagnostic remains within its numerical limit, while formal Gate 7 is pending. They do *not* support the preregistered claim of a five-point return-phase advantage over the stronger scan baseline.

2 Extension design

2.1 State and online clustering

Each admitted entry stores a topic, a fine-grained semantic cell, insertion and last-hit ticks, and decayed hit mass. Topics retain normalized centroids, resident IDs, demand, and miss mass. Cells retain centroids, residents, support, and last observation. The bounds are $\min(64, C)$ topics and $2C$ cells for capacity C .

On a miss, the normalized embedding joins the highest-cosine topic above `topic_threshold`, or creates a bounded topic. The same rule assigns a cell using `cell_threshold`. Centroids use a normalized exponential moving average. Ties always choose the lowest stable state ID. Low-support empty cells remain temporarily as bounded ghost history, allowing a second similar miss to challenge an incumbent even if the first candidate was rejected.

One logical tick occurs per cache hit or inserted miss. A statistic x last updated at t_0 decays lazily as

$$x(t) = x(t_0) 2^{-(t-t_0)/H},$$

where H is `demand_half_life`; $H = \infty$ disables decay. Lazy decay keeps the common hit path constant-time.

2.2 Quotas, value, and admission

For topic c , recent miss pressure and quota weight are

$$p_c = \frac{m_c + 1}{d_c + 2}, \quad w_c = (d_c p_c + \epsilon)^q,$$

where m_c is miss mass, d_c is demand, and q is `quota_strength`. Largest-remainder apportionment produces nonnegative integer quotas that sum exactly to capacity. A resident e in cell s has retention value

$$V_e = \frac{\text{support}(s)}{\max(1, |s|)} + 0.25 \text{ hitmass}(e).$$

The first term rewards demonstrated semantic coverage and discounts redundant residents in the same cell.

Below capacity, every valid candidate is admitted. At capacity, a candidate is rejected when its decayed cell support is below 1.5 or its topic quota is zero. Otherwise it competes against the weakest eligible resident, either inside its topic or in the most over-quota donor topic. Admission requires candidate value at least 1.05 times victim value. Victim ties use value, last hit, insertion age, and stable ID representation in that order.

2.3 Complexity and intended trade-off

Hits are $O(1)$ apart from periodic quota refresh. Miss assignment is $O((T + S)D)$ for bounded topics T , cells S , and dimension D ; victim selection is $O(C)$. CARMA intentionally pays more policy CPU than LRU/LFU in exchange for semantic coverage. It should help under phase change and pollution, but need not dominate when a simple baseline already retains the working set.

3 Correctness and GPTCache integration

CARMA is opt-in; existing callers and the default LRU path are unchanged. The factory accepts `policy="CARMA"` and binds the storage callback internally, avoiding a subtle failure mode in which a prebuilt custom eviction instance is not connected to `SSDataManager._clear`. Insertions pass normalized embeddings and persisted metadata through a backward-compatible hook.

Invariant	Enforcement and test evidence
Count bound	Integer quotas sum to C ; randomized property sequences assert no resident overflow.
Scalar/vector agreement	Victims are deleted from FAISS before scalar commit; targeted rollback preserves earlier tombstones and fails closed when reconciliation cannot succeed.
Exactly-once callback	Batch victims are deduplicated and filtered against final residents before the out-of-lock callback.
Malformed metadata	Missing, non-finite, zero-length, or wrong-dimensional embeddings fail safely at full capacity.
Restart	Existing rows are reconstructed with a read-only scalar peek; demand/hit mass intentionally relearn from zero.
Determinism	Topic, cell, quota, donor, victim, and ghost-pruning ties each have a documented stable ordering.

Table 1: Core safety properties exercised by unit, property, failure-path, and SQLite/FAISS integration tests.

The failure-path suite includes callback exceptions, unhashable and representation-colliding IDs, re-admission within one batch, vector-delete failure after scalar soft deletion, and close/reopen reconstruction. The implementation becomes explicitly unhealthy after unrecoverable cleanup rather than continuing with split internal/external state.

3.1 Validation layers

The project uses four layers: (1) focused CARMA unit/property tests; (2) relevant upstream GPTCache eviction and SQLite/FAISS tests; (3) deterministic CI traces with expected schemas/digests; and (4) real SQLite plus FAISS runs in fresh child processes. Python 3.12.13 is the primary pinned target; an explicit Python 3.8 compatibility slice protects GPTCache’s older supported syntax. No optional backend may trigger a runtime package installation during verification.

3.2 Restart limitation

Resident semantic structure is reconstructed from stored embeddings, but adaptive demand and hit mass are not persisted. They restart at zero and relearn online. Persisting them correctly would require a versioned metadata schema and migration rules across GPTCache backends; this is future work, not silently simulated state.

4 Experimental setup

The protocol was frozen before the inferential test. Validation and test seeds are disjoint, and test results never enter parameter selection. The 96-point grid crosses topic thresholds $\{.70, .80, .90\}$, cell thresholds $\{.95, .97\}$, half-lives $\{100, 500, 2000, \infty\}$, and quota strengths $\{0, .5, 1, 2\}$. Three validation seeds and three workloads select the highest workload-normalized valid-hit rate subject to zero synthetic false hits. The selected configuration is $(.70, .97, 500, 1.0)$.

Workload	Frozen construction	Purpose
Stationary Zipf	10,000 requests, exponent 1.2, 15% one-shot	Steady skew and non-regression.
Category shift	Five 2,000-request phases; 80% demand moves	Adaptation to new hot topics and recovery lag.
Pollution scan	3,000 warm, 4,000 unique, 3,000 return	Scan resistance and working-set recovery.
MOSS novel-long	Longest quartile, token-bucket stratified, 200 unique	Expected-miss overhead and recorded-response/token accounting.
QQP safety	Component-disjoint calibration/test split	Human-labeled semantic precision and false-hit control.

Table 2: Primary and safety workloads. Synthetic primary runs use capacity 100 and ten paired seeds; real system runs use five fresh processes per policy (15 policy processes total).

A hit is *valid* only if the returned answer ID is allowed by the request’s ground-truth concept. False hits save zero valid tokens. Primary metrics are valid-hit rate (VHR), hit precision, false-hit rate, opportunity recall, and safe token-saving ratio. System metrics include request latency percentiles, throughput, CPU time, sampled RSS, and storage invariants.

Paired deltas use 10,000-resample seed bootstraps. Confirmatory VHR comparisons against LRU, LFU, and the per-seed stronger baseline share an exact Wilcoxon Holm family. Capacity sweeps use the same capacity-100 traces at capacities 20, 50, 100, and 200. Ablations disable clustering, decay, admission, or quota-aware eviction separately.

The compact synthetic catalog preserves cosine 0.96 within a cell, 0.88 across cells in one topic, 0.72 across topic-local concepts, and at most 0.12 across topics. It makes the 10,000-request CPU grid tractable while remaining a controlled model, not a natural-language claim.

5 Synthetic results

Workload	LRU VHR	LFU VHR	CARMA VHR	CARMA vs. stronger
Stationary	43.457%	47.514%	54.636%	+7.122 pp
Category shift	63.376%	52.181%	68.261%	+4.885 pp
Pollution scan (whole)	58.400%	59.198%	59.200%	+0.002 pp
Pollution return phase	97.333%	99.993%	100.000%	+0.0067 pp

Table 3: Ten-seed means at capacity 100. “Stronger” is chosen per seed and metric before forming paired differences.

Figure 1 summarizes the primary whole-trace comparisons.

Valid-hit-rate delta vs stronger baseline

Paired seed mean and 95% bootstrap CI; primary capacity 100.
inferential full protocol.

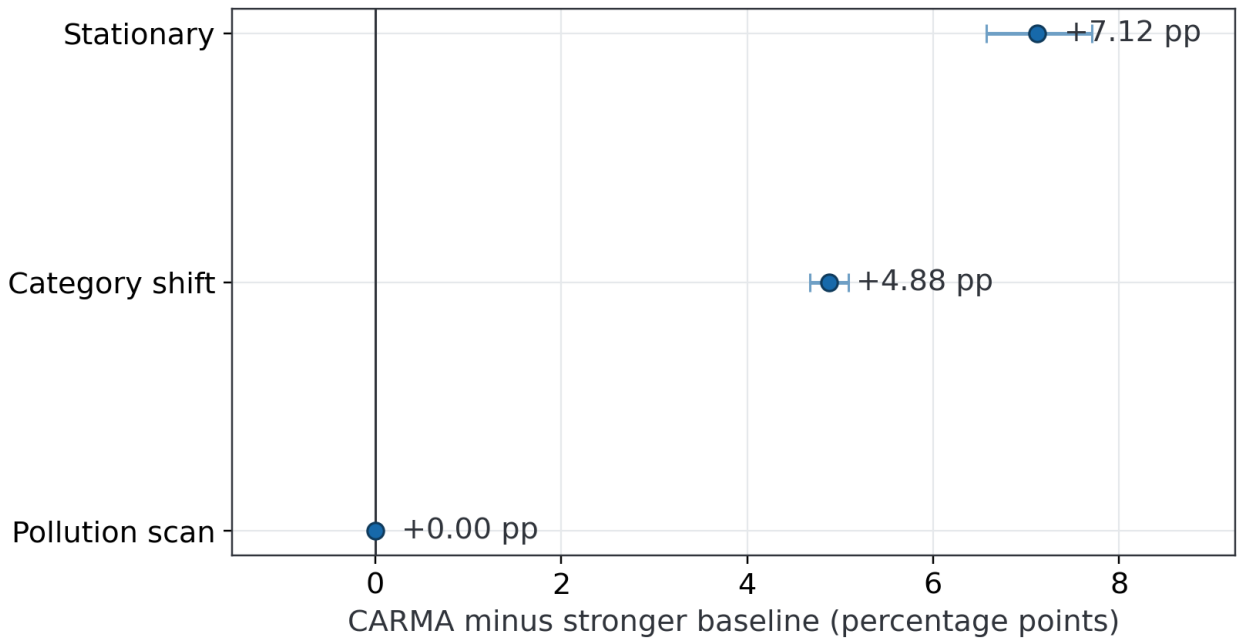


Figure 1: Primary whole-trace VHR difference between CARMA and the per-seed stronger LRU/LFU baseline. Error bars are paired 95% bootstrap intervals.

The category-shift result passes its preregistered gate: +4.885 pp, CI [4.677, 5.088], Holm-adjusted $p = 0.0234$. Stationary performance also passes the non-regression gate with CI lower bound +6.572 pp. Safe token-saving deltas pass in all workloads: stationary +9.503 pp [8.641, 10.443], shift +5.251 pp [5.009, 5.475], and pollution +0.0015 pp [0, 0.0037].

Figure 2 makes the failed margin visible against the frozen threshold.

This is a substantive failed hypothesis, not a measurement failure. The trace demonstrates that CARMA protects the original set, but LFU protects it almost equally well. The paired delta is +0.0067 pp, CI [0, 0.0167], Holm $p = 1.0$. In the separate exploratory 0.95-threshold hard-negative diagnostic, every policy produced one deliberate false hit per 200 requests (0.5% FHR), so the CARMA–baseline false-hit difference was zero. That diagnostic tests accounting and does not enter configuration selection or the primary threshold-0.97 claim.

Pollution scan return-phase valid-hit-rate delta

Paired seed mean and 95% bootstrap CI; primary capacity 100.
inferential full protocol.

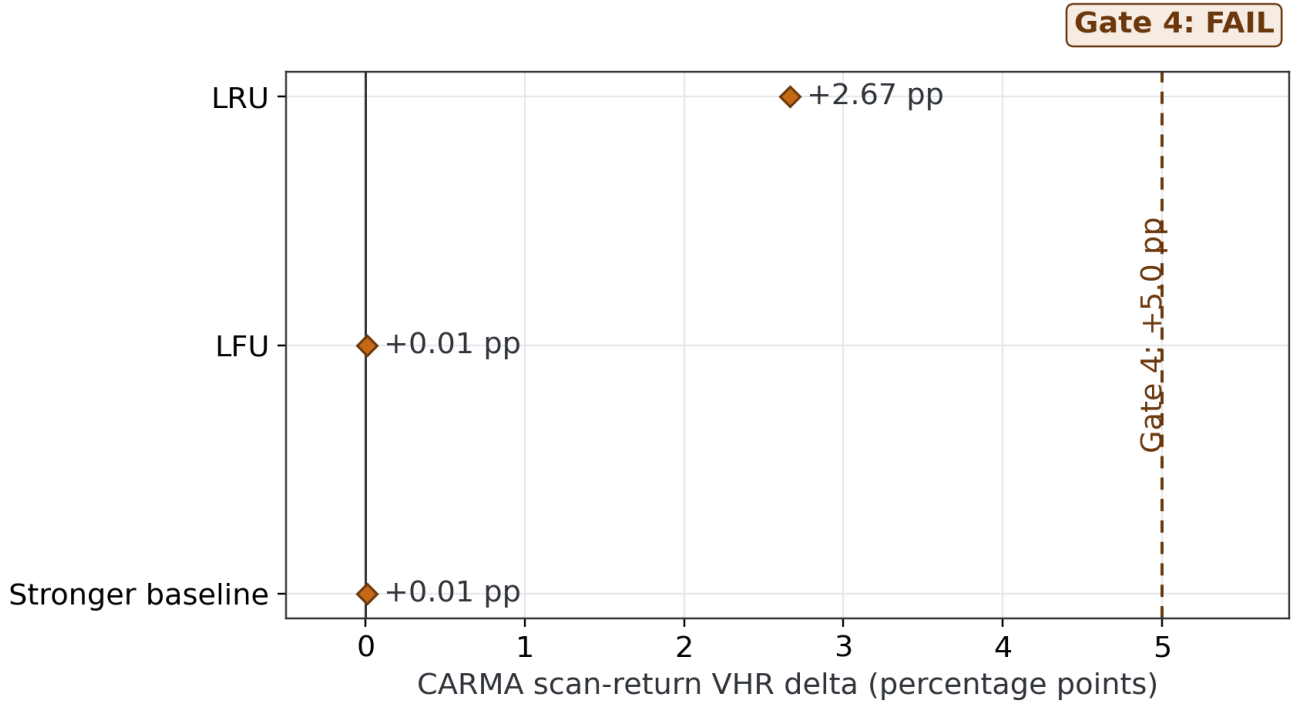


Figure 2: Return-phase scan comparison. CARMA is perfect, but LFU misses only two of 30,000 requests; the stronger-baseline delta is far below the preregistered +5.0 pp gate.

6 Sensitivity and ablations

Figure 3 shows the exploratory fixed-trace capacity sweep.

CARMA’s advantage is largest when capacity is constrained and under category shift. Averaging only for descriptive orientation across the three workloads, its VHR is 43.49%, 54.35%, and 63.97% at capacities 20, 50, and 200, respectively, compared with LRU’s 33.16%, 46.41%, and 60.74%.

Policy	Stationary	Category shift	Pollution scan
CARMA	54.636%	68.261%	59.200%
No clustering	54.983%	67.874%	59.200%
No decay	55.108%	59.980%	59.200%
No admission	53.025%	67.850%	55.388%
No quota eviction	54.983%	67.874%	59.200%

Table 4: Ablation mean VHR at capacity 100. These are descriptive component checks, not an additional confirmatory family.

Decay is the clearest adaptive component: removing it loses 8.281 points on category shift while slightly helping stationary traffic. Admission matters most on the scan workload; disabling it loses 3.812 points. In this synthetic geometry, no-clustering and no-quota produce identical observable hit/eviction behavior despite different topic counts. Their causal effects are therefore not separately identifiable here and should not be over-interpreted.

Recovery-lag evidence is weak: CARMA averages 1,808 requests versus the 2,000-request censoring limit for both baselines. Yet 31/40 CARMA and all 40 baseline transitions are censored; exploratory exact Wilcoxon $p = 0.0625$.

Valid-hit rate across cache capacities

95% bootstrap CIs; fixed capacity-100 traces. Capacity 100 uses 10 seeds. Exploratory sweep points use 5; inferential full protocol.

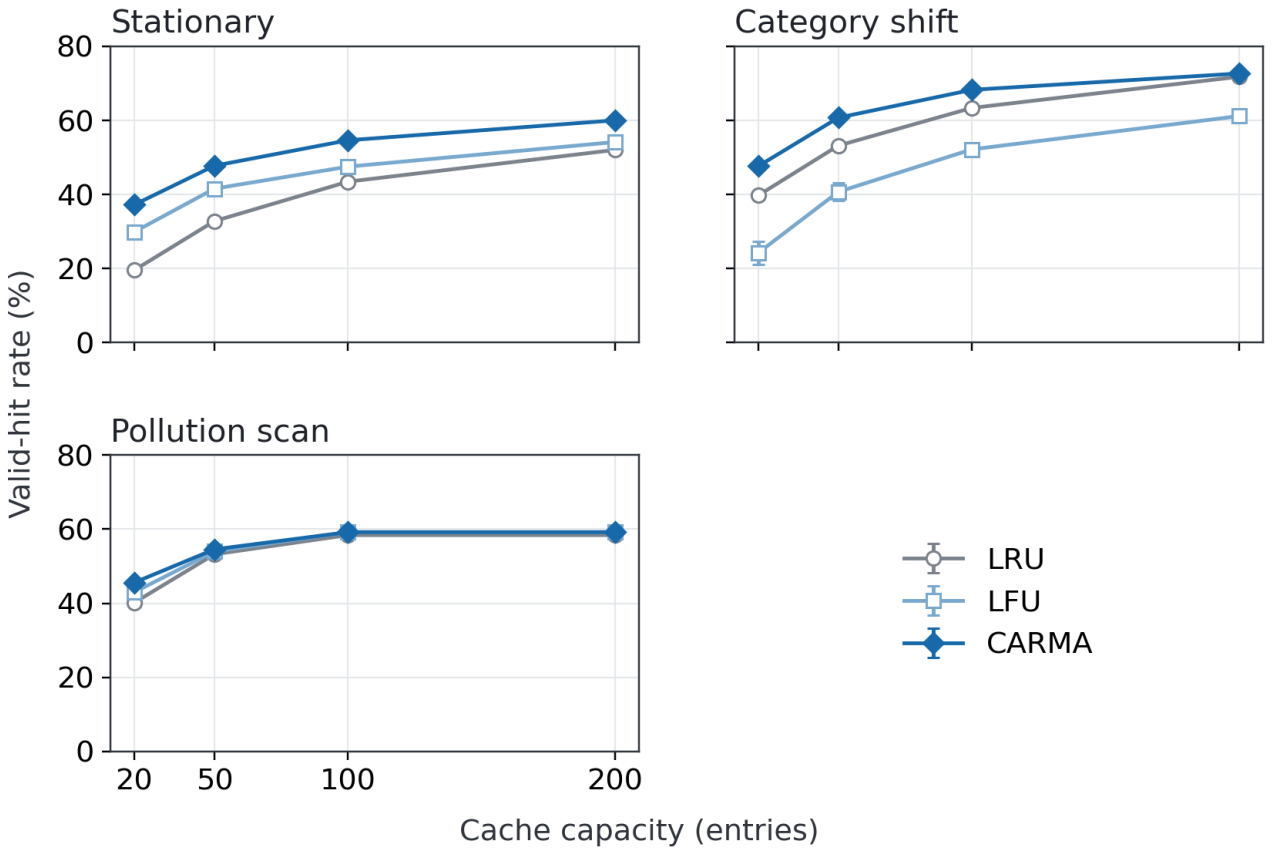


Figure 3: Valid-hit rate on identical traces while cache capacity changes. Capacity 100 uses ten seeds; other exploratory points use five.

7 Real-system, MOSS, and semantic-safety evidence

The system benchmark uses the actual GPTCache `SSDataManager`, `SQLite`, and `FAISS` path, but replays precomputed vectors from the controlled synthetic catalog rather than running `paraphrase-albert-onnx`. Each policy runs in a fresh process on the same 3,000-request pollution trace, with immediate physical deletion and a final round-trip check of every active vector. Across five seeds, every run has zero false hits, zero stale candidates, and equal final scalar/vector counts.

Figure 4 separates policy-only work from post-embedding cache-path cost.

SQLite/FAISS latency and resource measurements

Precomputed vectors; embedding excluded. Descriptive per-seed means.
n=5 seed(s), mode=full; no across-seed Gate 7 rule is frozen.

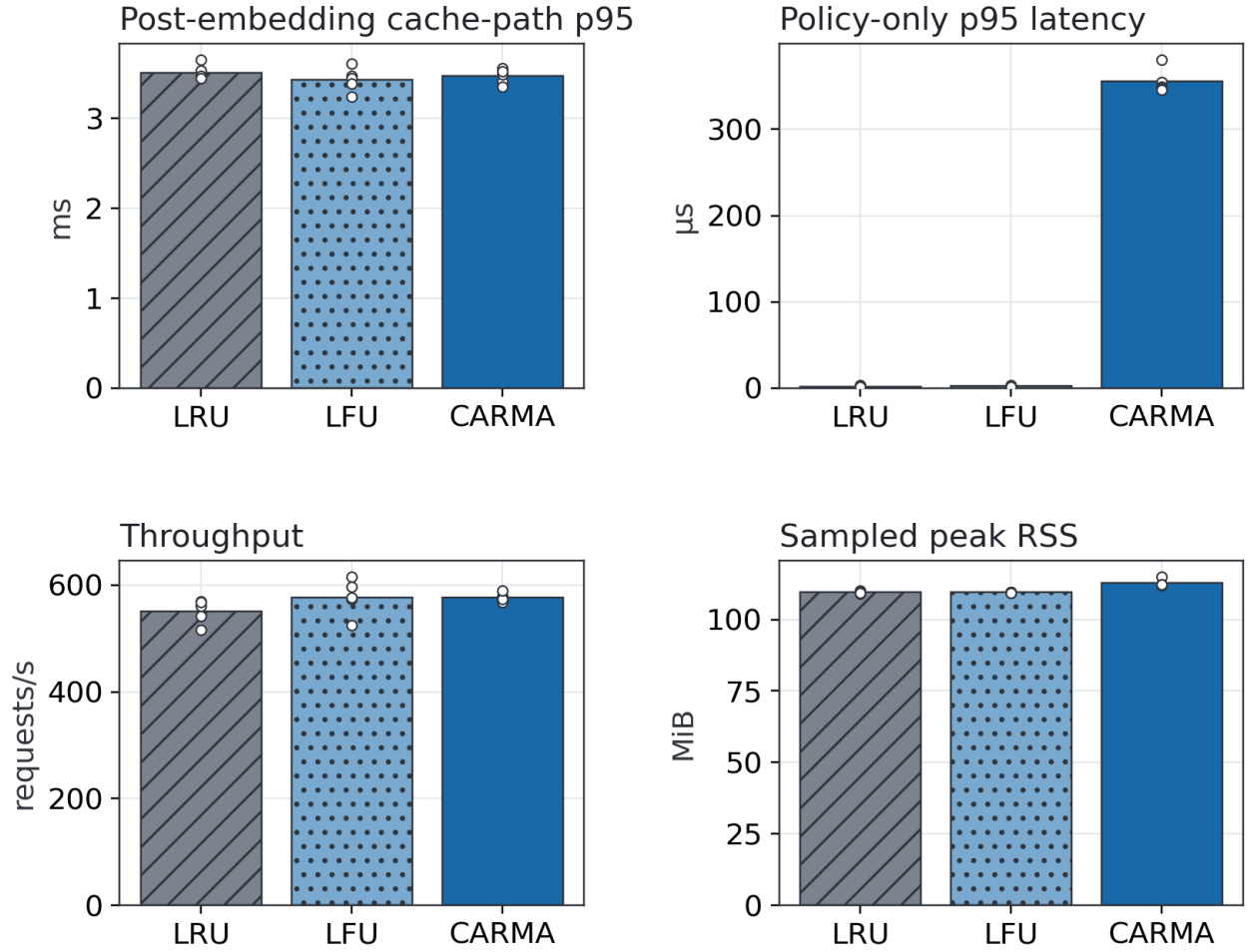


Figure 4: Real SQLite/FAISS measurements using precomputed synthetic vectors. Bars are five-seed means and circles are individual fresh-process runs. Policy-only overhead is shown separately from post-embedding cache-path request cost; embedding generation is excluded.

Policy	Mean	p50	p95	p99
LRU	1.804 ms	1.032 ms	3.511 ms	4.602 ms
LFU	1.724 ms	0.953 ms	3.433 ms	4.181 ms
CARMA	1.723 ms	0.875 ms	3.469 ms	3.952 ms

Table 5: Five-seed means of each fresh process’s request-latency summaries. The measured scope is the post-embedding cache path using precomputed vectors.

CARMA adds roughly 354 μ s to policy-only p95, reflecting bounded cosine assignment and victim selection. Post-embedding cache-path p95 nevertheless ranges from $0.936\times$ to $1.032\times$ LRU because policies create different hit, admission, and storage-cleanup mixes. The worst ratio across seeds is $0.997\times$ LRU for throughput and $1.043\times$ LRU (+4.8 MiB) for sampled peak RSS. Mean one-core CPU utilization is 96.9% for CARMA versus 94.1% for LRU. GPU use is not applicable to this CPU-only path, and macOS did not expose portable process I/O byte counters. Every seed satisfies the individual numerical checks; formal Gate 7 remains pending because the frozen contract omitted an across-seed aggregation rule and specified an ONNX integration embedding that this run did not use.

Only each run’s mean and quantiles were retained; individual request-latency samples were not retained in the integration artifacts. Figure 4 therefore shows the cross-seed distribution of p95 summaries, not a reconstructable per-request CDF. This is an artifact-granularity limitation despite collecting the frozen mean, p50, p95, and p99 metrics in process.

7.1 Pinned MOSS recorded-response replay

The CC-BY-4.0 MOSS archive [3] is fixed by repository revision and SHA-256. A full streaming CRC-verified pass covers 301,332 source conversations and 823,634 actual turns, then retains the lowest stable-hash 2,048-turn sample. The 200-request longest-quartile trace contains 557,243 context, prompt, and answer tokens. All requests are unique by design, so all 200 are expected misses and the exact recorded MOSS answer is replayed; live model calls and false hits are both zero. Two runs produce byte-identical manifests, pool, requests, and CSV.

The real archive also exposed 30,000 rows with `conversation_id=-1` and 2,727 declared-turn-count mismatches. Before producing a result, the parser was amended to hash canonical rows for negative sentinel IDs and to treat contiguous chat keys as authoritative while retaining both declared and actual counts. The final manifest records these anomalies and preserves the separate prepare manifest.

7.2 Held-out QQP safety

Final report build note: this paragraph and the gate table are filled from the checksum-linked QQP `result.json` after the pinned ONNX embedding, calibration-only threshold selection, and one-shot component-disjoint test. No synthetic result may select or alter the QQP threshold.

Gate	Status	Current adjudication
1	PENDING	Clean final verifier artifact not yet linked.
2	PENDING	QQP embedding/evaluation in progress; frozen FHR scope is ambiguous.
3	PASS	Shift VHR +4.885 pp; CI [4.677, 5.088]; Holm $p = .0234$.
4	FAIL	Return VHR +0.0067 pp versus required +5.0 pp.
5	PASS	Stationary VHR delta CI lower bound +6.572 pp.
6	PASS	Safe-token delta CI lower bound meets every workload limit.
7	PENDING	Every precomputed-vector seed meets the diagnostics, but no frozen aggregation rule and no ONNX-path run.
8	PENDING	Paired clean-container evidence not yet linked.

Table 6: Success-gate status reconciled to the machine-readable audit. Pending evidence is not counted as a pass, and the failed preregistered gate is retained.

8 Discussion, protocol audit, and limitations

8.1 What the experiments establish

CARMA’s strongest evidence is phase adaptation: decay prevents obsolete lifetime popularity from dominating after a topic shift, while topic quotas preserve coverage. Admission also protects scan performance. The implementation cost is measurable in policy-only latency; every tested seed stays inside the diagnostic numerical limits on this CPU/SQLite/FAISS path, but the formal system gate remains pending.

The scan gate explains why raw perfection is not enough. CARMA achieves 100% return VHR, but the correct comparator is the stronger paired baseline, not LRU alone. LFU is already 99.993%; the prespecified five-point margin is impossible to claim from this trace. A future scan workload should make the hot set large enough, or interleave enough recurrent cold items, to distinguish coverage policies without changing the gate after observation.

8.2 Post-execution protocol conformance

The full bundle’s hashes, seed pairing, selected configuration, 480 aggregate rows, bootstraps, Wilcoxon tests, and Holm adjustment were independently recomputed exactly. The audit also found five deviations that this report does not hide:

1. the historical full bundle contains summary CSVs/metadata but not the promised per-request and process-resource JSONL, limiting reconstruction of individual phase events from that bundle alone;
2. synthetic policy execution order is fixed rather than randomized; state is isolated and timing is disabled, so this is unlikely to bias VHR;
3. synthetic false-hit intervals are seed bootstraps, not Wilson bounds; they must not be cited as the substantive semantic-safety interval; and
4. the preregistered scale sweep and secondary negative/ceiling controls were not run. Capacity sweeps and the four component ablations were run; and
5. the integration runs used precomputed 2,982-dimensional synthetic vectors rather than the frozen ONNX embedding, so their timing covers only the post-embedding cache path and Gate 7 remains diagnostic.

The exact historical result remains tied to clean commit `28129f0b9785232827e741c0e2ff2bb96cc19423`; later tooling changes cannot retroactively alter its status. A separate deviation ledger documents impact and future-run remediation.

8.3 External validity

Synthetic vectors isolate eviction behavior but are not natural language. QQP tests pairwise semantic matching, not conversational distribution shift. MOSS is an exact-key recorded-response negative control rather than a CARMA semantic-policy comparison. System timing comes from one Apple-arm64 machine, five seeds, one synthetic embedding geometry, capacity 100, and SQLite/FAISS; embedding generation is excluded. Restart loses adaptive counters. These boundaries limit generalization but make each claim traceable.

9 Conclusion and future work

This project delivers a cleanly factored GPTCache feature branch, an online cluster-aware policy, robust cross-store deletion, focused and upstream tests, frozen paired experiments, real backend measurements, pinned natural-data checks, deterministic analysis, container/CI definitions, and an auditable report. The central hypothesis is only partly supported: CARMA is compelling under category shift and passes the synthetic stationary VHR non-regression gate, while its preregistered scan advantage over the strongest baseline fails.

The most valuable next steps are to persist versioned adaptive state, replace linear victim scans with per-topic heaps, test more embedding models and cache sizes on Linux, design a non-saturated pollution workload before collecting new results, and separate quota from clustering effects with traces whose topic pressures actually diverge. Any new workload or gate must be frozen before looking at its test outcomes.

Lecturer-criterion audit

Criterion	Status	Evidence
Correctness (40%)	PENDING	Unit/property/failure/integration and upstream slices are implemented; clean final verifier evidence is pending and will be retained under <code>artifacts/samples/verification/</code> .
Reproducibility (30%)	PENDING	Exact pins, Dockerfile, immutable CI actions, and deterministic CI/MOSS/analyzer outputs are implemented; paired clean-container evidence is pending.
Performance (15%)	Mixed	Gates 3, 5, 6 pass; each system seed meets the diagnostic checks but Gate 7 is pending; Gate 4 fails.
Clarity (15%)	Complete	One-page baseline rationale, policy/protocol docs, README commands, labeled figures, limitations, and this report.

Appendix: Artifact map

Artifact	Purpose
<code>gptcache/manager/</code>	<code>eviction/carma.py</code> implements CARMA; <code>factory.py</code> , <code>data_manager.py</code> , and <code>eviction_manager.py</code> provide wiring and cleanup.
<code>tests/unit_tests/eviction/test_carma.py</code> ; <code>tests/project_tests/</code> <code>docs/project/</code>	Unit, property, failure-path, protocol, analyzer, and real SQLite/FAISS tests. <code>baseline-justification.md</code> , <code>baseline-validation.md</code> , <code>policy-design.md</code> , <code>experiment-contract.md</code> , <code>protocol-deviations.md</code> , <code>reproducibility.md</code> , and <code>chart-map.md</code> .
<code>benchmarks/carma/</code> ; <code>scripts/</code>	Synthetic, integration, QQP, MOSS, statistics, and analysis; benchmark, analyzer, and verifier wrappers.
<code>Dockerfile.project</code> ; <code>.dockerignore</code> ; <code>requirements-*.lock</code> ; <code>scripts/generate_hashed_locks.py</code> <code>README.md</code> ; <code>.github/workflows/carma-ci.yml</code>	Pinned container, allowlisted context, hashed dependency locks, and lock generator.
<code>artifacts/samples/</code>	Installation, benchmark how-to, and immutable host/container CI. <code>full/</code> , <code>integration/</code> , <code>moss/</code> , and <code>qqp/</code> hold curated raw evidence; <code>analysis/</code> , <code>verification/</code> , <code>README.md</code> , and <code>SHA256SUMS</code> hold the audit, figures, checks, inventory, and hashes. Full source archives, model files, and embeddings are excluded.
<code>docs/project/completion-audit.md</code> ; <code>docs/project/draft-pr.md</code> ; <code>docs/project/report.pdf</code>	Rubric map, unpublished upstream PR draft, and this final single-PDF handoff.

References

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